Building an Interactive Storymap using {mapgl}

A Visual Journey Through My Life.

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1 Introduction

Interactive storymaps have become an increasingly popular tool for data visualization and storytelling, particularly in fields like urban planning, transportation analysis, and geographic data science. They allow us to combine spatial data with narrative elements, creating compelling visual stories that engage audiences while conveying complex information effectively.

In this post, I'll walk you through creating an interactive storymap using the {mapgl} R package —a powerful tool that leverages MapLibre GL JS to create stunning, performant web maps. This particular storymap traces my personal journey across different locations where I've lived, studied, and worked, from Nepal to the United States.

2 Why {mapgl} for Storymaps?

The {mapgl} package offers several advantages for transportation planners and data analysts:

- **Performance**: Built on MapLibre GL JS, it handles large datasets and complex visualizations smoothly
- **Interactivity**: Native support for user interactions, animations, and dynamic content
- **Customization**: Extensive styling options and the ability to use custom map tiles
- **Integration**: Seamless integration with Shiny for reactive web applications
- **Open Source**: No API keys required when using open map styles

3 Interactive Storymap Demo

Below is the complete interactive storymap. You can scroll through the different sections to see how the map smoothly transitions between locations, each representing a significant milestone in my journey.

```
#| '!! shinylive warning !!': |
#|   shinylive does not work in self-contained HTML documents.
#|   Please set `embed-resources: false` in your metadata.
#| standalone: true
#| viewerHeight: 600

source("app.R")
```

4 Understanding the Code Structure

The storymap application follows a typical Shiny structure with some specialized {mapgl} components. Following is the source code for this dashboard (Please click on the numbers on the far right for additional explanation of syntax and mechanics):

```
library(shiny)
library(mapgl)

ui <- shiny::fluidPage(
  tags$link(
    href = "https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600&
display=swap",
    rel="stylesheet"
  ),
  mapgl::story_map(
    map_id = "map",
    font_family = "Poppins",
    sections = list(
      "intro" = mapgl::story_section(
        title = "Introduction",
        content = "Hello I am Pukar Bhandari; and this is my life journey."
      ),
      "birth_place" = mapgl::story_section(
        title = "Birth Place",
        content = "This is where I was born."
      ),
      "lower_primary" = mapgl::story_section(
        title = "Kankai Education Foundation",
        content = "This is where I studied Nursery to Grade 3."
      ),
      "upper_primary" = mapgl::story_section(
        title = "Laligurans English School",
        content = "This is where I studied Grade 4 & Grade 5."
      ),
      "middle_school" = mapgl::story_section(
        title = "Birat Jyoti English Secondary School",
        content = "This is where I studied Grade 6 & Grade 7."
      ),
```

```
      "secondary_school" = mapgl::story_section(
        title = "Little Flowers' English School",
        content = "This is where I studied Grade 8 to Grade 10."
      ),
      "higher_secondary" = mapgl::story_section(
        title = "Kanchanjunga English Higher Secondary School",
        content = "This is where I studied Higher Secondary (Plus 2)."
      ),
      "bachelors" = mapgl::story_section(
        title = "Institute of Engineering Pulchowk Campus",
        content = "This is where I obtained my Bachelor's Degree in
Architecture."
      ),
      "masters" = mapgl::story_section(
        title = "University of Utah",
        content = "This is where I obtained my Master of City & Metropolitan
Planning degree."
      ),
      "work_1" = mapgl::story_section(
        title = "Metro Analytics",
        content = "This is where I lived and worked between 2023-2025."
      ),
      "work_2" = mapgl::story_section(
        title = "Wasatch Front Regional Council",
        content = "This is where I live and work currently."
      )
    ),
    map_type = "maplibre"
  )
)

server <- function(input, output, session) {
  output$map <- mapgl::renderMaplibre({
    mapgl::maplibre(
      style = "https://tiles.openfreemap.org/styles/liberty",
      center = c(0, 0),
      zoom = 2.75,
      scrollZoom = FALSE
    ) |>
      mapgl::set_projection(projection = "globe") |>
      mapgl::add_globe_control() |>
      mapgl::add_navigation_control(visualize_pitch = TRUE) |>
      mapgl::add_globe_minimap(position = "bottom-left")
  })

  mapgl::on_section("map", "intro", {
    mapgl::maplibre_proxy("map") |>
      mapgl::clear_markers() |>
```

```r
    mapgl::fly_to(
      center = c(0, 0),
      zoom = 2.75,
      pitch = 0,
      bearing = 0
    )
})

mapgl::on_section("map", "birth_place", {
  mapgl::maplibre_proxy("map") |>
    mapgl::fly_to(center = c(87.99371750247397, 26.646533355308083),
           zoom = 16,
           pitch = 49,
           bearing = 12.8)
})

mapgl::on_section("map", "lower_primary", {
  mapgl::maplibre_proxy("map") |>
    mapgl::fly_to(center = c(87.99734666704784, 26.646179601882675),
           zoom = 16,
           pitch = 49,
           bearing = 12.8)
})

mapgl::on_section("map", "upper_primary", {
  mapgl::maplibre_proxy("map") |>
    mapgl::fly_to(center = c(88.01116125198942, 26.608294715049524),
           zoom = 16,
           pitch = 49,
           bearing = 12.8)
})

mapgl::on_section("map", "middle_school", {
  mapgl::maplibre_proxy("map") |>
    mapgl::fly_to(center = c(87.99218594227516, 26.620735015314324),
           zoom = 16,
           pitch = 49,
           bearing = 12.8)
})

mapgl::on_section("map", "secondary_school", {
  mapgl::maplibre_proxy("map") |>
    mapgl::fly_to(center = c(87.99006631463702, 26.63185913816036),
           zoom = 16,
           pitch = 49,
           bearing = 12.8)
})
```

```r
  mapgl::on_section("map", "higher_secondary", {
    mapgl::maplibre_proxy("map") |>
      mapgl::fly_to(center = c(87.99921328178023, 26.633744364809765),
             zoom = 16,
             pitch = 49,
             bearing = 12.8)
  })

  mapgl::on_section("map", "bachelors", {
    mapgl::maplibre_proxy("map") |>
      mapgl::fly_to(center = c(85.31840215760691, 27.681136558618093),
             zoom = 16,
             pitch = 49,
             bearing = 12.8)
  })

  mapgl::on_section("map", "masters", {
    mapgl::maplibre_proxy("map") |>
      mapgl::fly_to(center = c(-111.84448004883544, 40.76106105996334),
             zoom = 16,
             pitch = 49,
             bearing = 12.8)
  })

  mapgl::on_section("map", "work_1", {
    mapgl::maplibre_proxy("map") |>
      mapgl::fly_to(center = c(-84.3938999520061, 33.76728402587881),
             zoom = 16,
             pitch = 49,
             bearing = 12.8)
  })

  mapgl::on_section("map", "work_2", {
    mapgl::maplibre_proxy("map") |>
      mapgl::fly_to(center = c(-111.90422445885304, 40.76973849954712),
             zoom = 16,
             pitch = 49,
             bearing = 12.8)
  })
}

shiny::shinyApp(ui, server)
```

Line 1

Load {shiny} package for building reactive web applications

Line 2

Load {mapgl} package for interactive mapping and storymap functionality

Line 4
: Define the user interface - what users see and interact with

Line 9
: Create the main storymap interface with sidebar and map coordination

Line 12
: Define each story section with title and descriptive content

Line 62
: Define the server function - handles user interactions and updates

Line 63
: Render the interactive MapLibre map with styling and controls

Line 65
: Use free OpenFreeMap tiles (no API key required)

Line 70
: Set up 3D globe projection for better geographic context

Line 76
: Handle section transitions - when user reaches each story step

Line 168
: Launch the Shiny application combining UI and server components